

BUSINESS PROCESS MANAGEMENT

IS 2206

Lecture 04

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OUTLINE

1. Why Process Modeling?
2. Modeling Levels
3. Business Process Model and Notation (BPMN)

TOPIC 01

Why Process Modeling?



1.1 What is Business Process Modelling?

Business Process Modelling is the graphical representation of a company's business processes or workflows, as a means of identifying potential improvements.

- This is usually done through different graphing methods, such as the flow chart.
- BP modelling used to map 2 different states of the process:
 - As-is, the state of the process as it is right now, without making any changes
 - To-be, the future state, after making the changes or the improvements

1.2 Why Process Modeling?

- For a better understanding of the existing business processes.
- To document and communicate business processes.
- The basis for improving existing business processes through As-is and To-be process models.
- The basis for testing and simulating new concepts.
- The basis for creating information technologies that support business processes.

1.3 BPM Standards

- Various BPM standards have been introduced for process development, but all of them have certain limitations.
- The latest version in use is the BPMN 2.0 standard. This standard supports both designing as well as programming language functionalities.

1.4 BPMN - Business Process Model and Notation

- Business Process Model and Notation (BPMN) is a standard for business process modeling that provides a graphical notation for specifying business processes in a Business Process Diagram (BPD), based on a flowcharting technique.
- BPMN has over 100 symbols. BPM suits give modeling facility with these symbols.
- BPMN provides a notation that can be readily understandable by all users:
 - from the business analyst who models the process conceptually
 - to the technical developers responsible for implementing the technology for the process.
 - to the people who will manage and monitor the process.

TOPIC 02

Modeling Levels



2.1 Level 1 – Descriptive BPMN

- A descriptive process model defines the scope of the end-to-end business process at the strategic level that managers can easily understand.
- The model serves the following purposes:
 - Clarify the scope of the end-to-end process
 - Identify resources and assign responsibilities
 - Identify KPIs of the business process
 - Review the process to determine improvement and/or automation.
- In this level the process model must be easy to understand even for people with no experience in BPMN.
- Usually, Level 1 diagrams contain only basic BPMN symbols

2.2 Level 2 - Analytical BPM

- Draws on the full panel of BPMN shapes and symbols, but the method emphasizes the diagram elements most commonly used.
- Analytical modeling, more detailed, showing all the steps, including the exception paths, required either to analyze process performance using simulation or to create detailed requirements for an IT implementation.
- Level 2 diagrams must be both valid according to the rules in the BPMN spec, and organized effectively as hierarchical representations of the end to end business process.
- Level 2 models are often used before software development starts.

2.3 Level 3 – Executable BPMN

- Executable modeling, where BPMN is part of the executable process implementation.
- While this capability of BPMN is a major reason for its widespread adoption, modeling at this level is somewhat vendor tool-dependent, as most BPM Suites do not support all of the gateway and event types defined in the BPMN spec, and may offer their own proprietary extensions as well.
- Level 3 diagrams thus typically impose additional validation constraints beyond those of the BPMN spec.

2.4 Modelling Tools

BizAgi

BizAgi Process Modeler
Modelling
BPMN
Freeware (bizagi.com)



- Bizagi Modeler downloadable links,
- <https://www.bizagi.com/en/platform/modeler>

Intalio

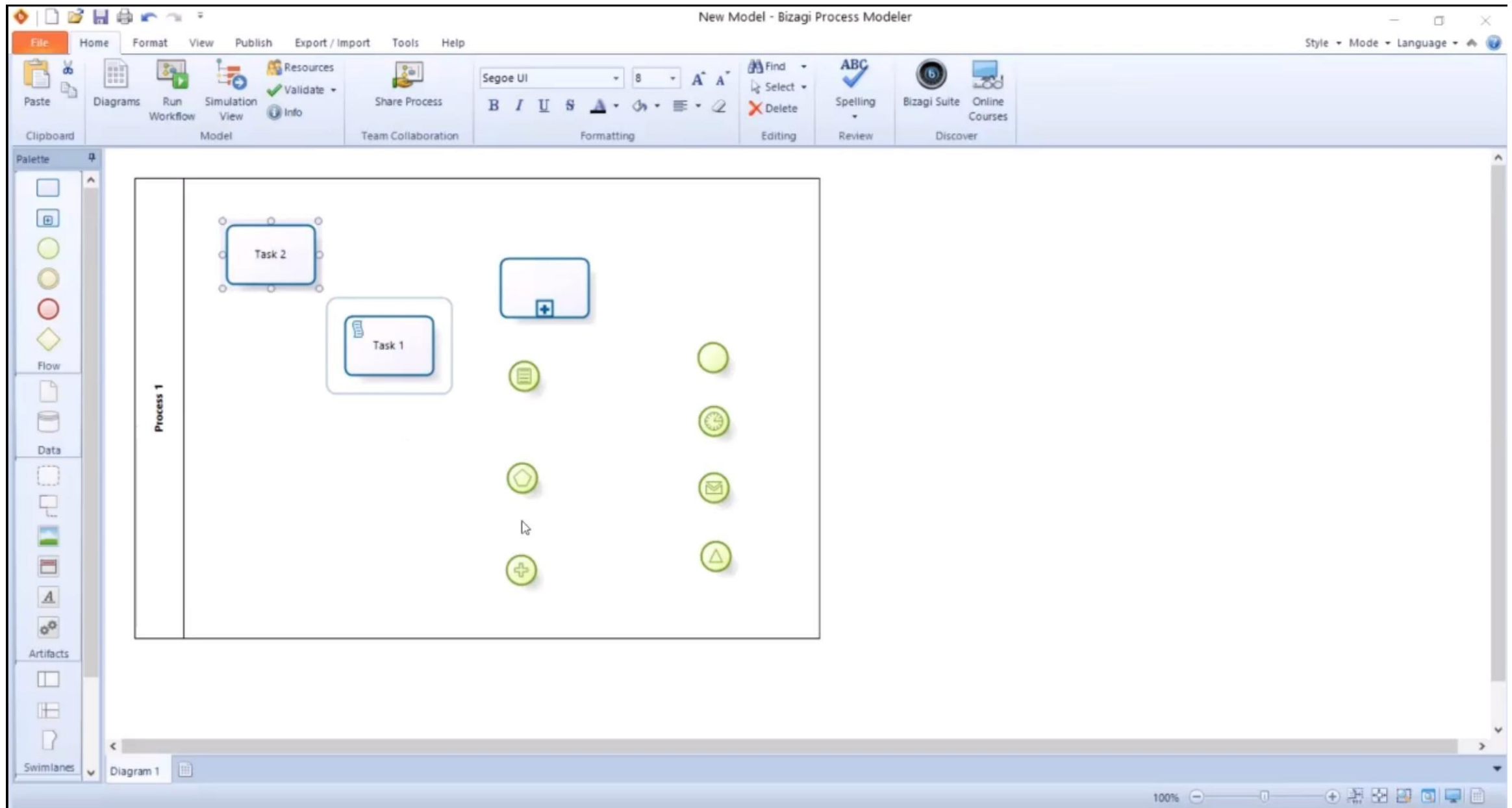
Designer, Server
Modelling, Execution
BPMN
Freeware (intalio.com)



IBM

Websphere
Modelling, Execution, Simulation
IBM Academic Initiative





TOPIC 03

Business Process Model and Notation (BPMN)



3.1 Private and Public Processes

Private Process

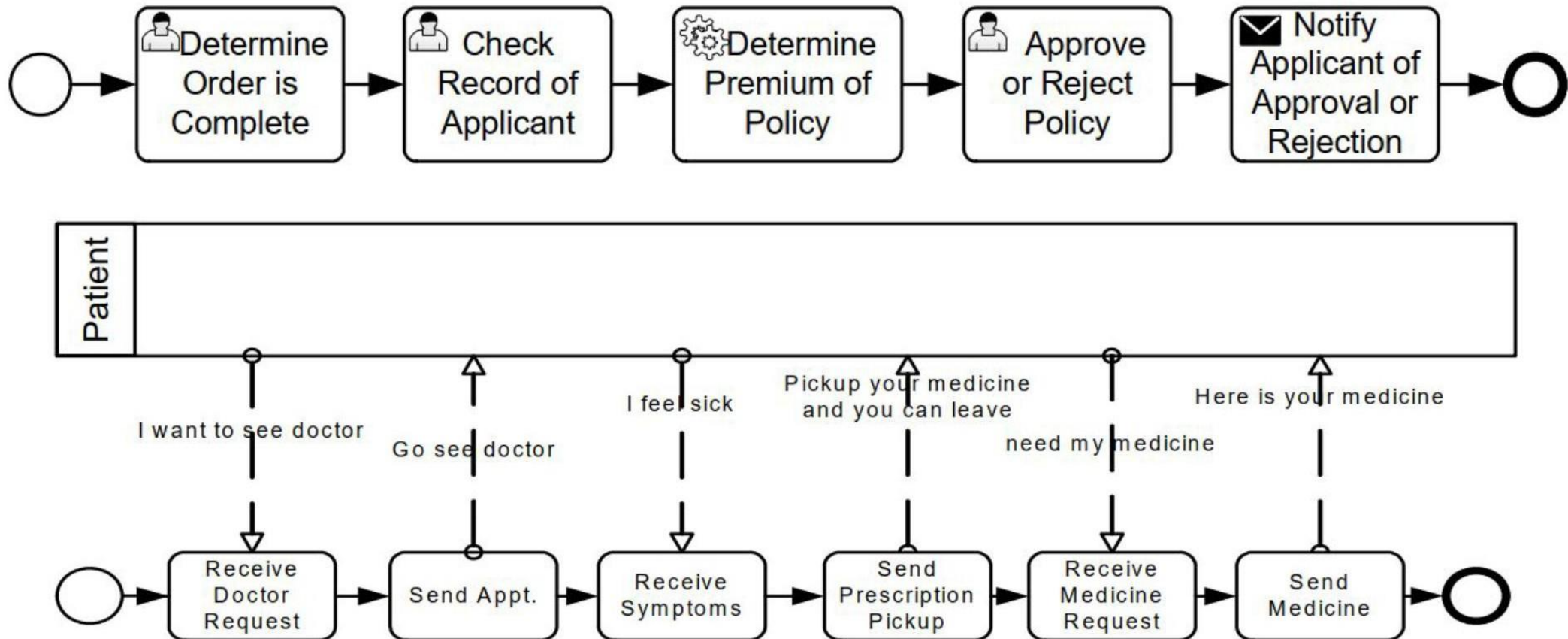
- Private Business Processes are those internal to a specific organization. These Processes have been generally called workflow or BPM Processes.
- A private Business Process will be contained within a single Pool. The Process flow is therefore contained within the Pool and cannot cross the boundaries of the Pool.

Public Process

- A public Process represents the interactions between a private Business Process and another Process or Participant .
- All other “internal” Activities of the private Business Process are not shown in the public Process. Thus, the public Process shows to the outside world the Message Flows and the order of those Message Flows that are needed to interact with that Process.
- Public Processes can be modeled separately or within a Collaboration to show the flow of Messages between the public Process Activities and other Participants.

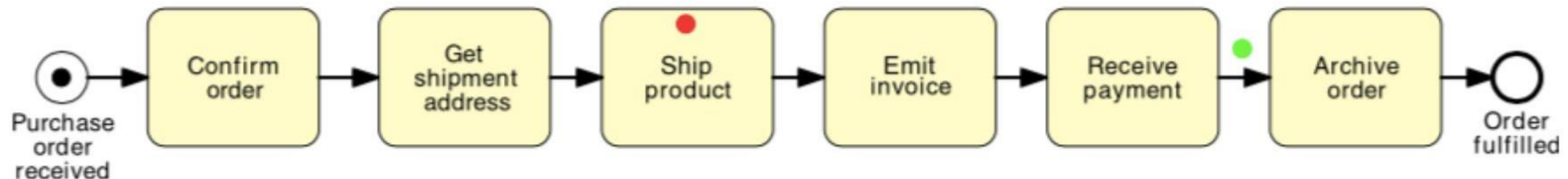
3.1 Private and Public Processes (Continued)

Example



3.2 General Guidelines

- Tokens - colored dots on top of a process model (Show the status of particular instance).
- Naming conventions
 - ✓ Activities label should begin with a verb followed by a noun. e.g. “Approve order”
 - ✓ Avoid labels with more than five words.
 - ✓ For events, the label should begin with a noun and end with a verb in past participle form.
 - ✓ Name of the process model should be a noun, preceded by an adjective, ex: “order fulfillment”.
 - ✓ We capitalize the first word of activity and event labels.
 - ✓ Do not capitalize the first word of process names, ex: the “order fulfillment” process.



3.2 General Guidelines (Continued)

Sequence Flow

- The sequence flow allows us to follow a transaction in a model.
- It is illustrated by a solid one-way arrow and is used to link **events**, **activities** and **gateways**.
- When modeling a business process, it is recommended that each event and activity have only one inbound and one outbound sequence flow to unambiguously illustrate what precedes and follows an activity or event.



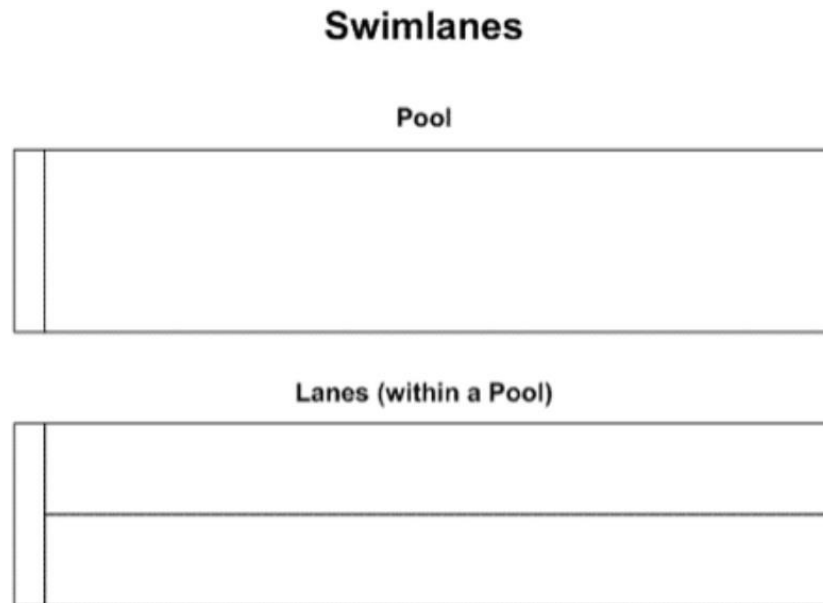
3.3 BPMN Elements

The Basic Categories of Elements are

1. Swim Lanes
 - Pools
 - Lanes
2. Flow Objects
 - Activities
 - Events
 - Gateways
3. Connecting Objects
4. Artifacts

3.3.1 Swim Lanes

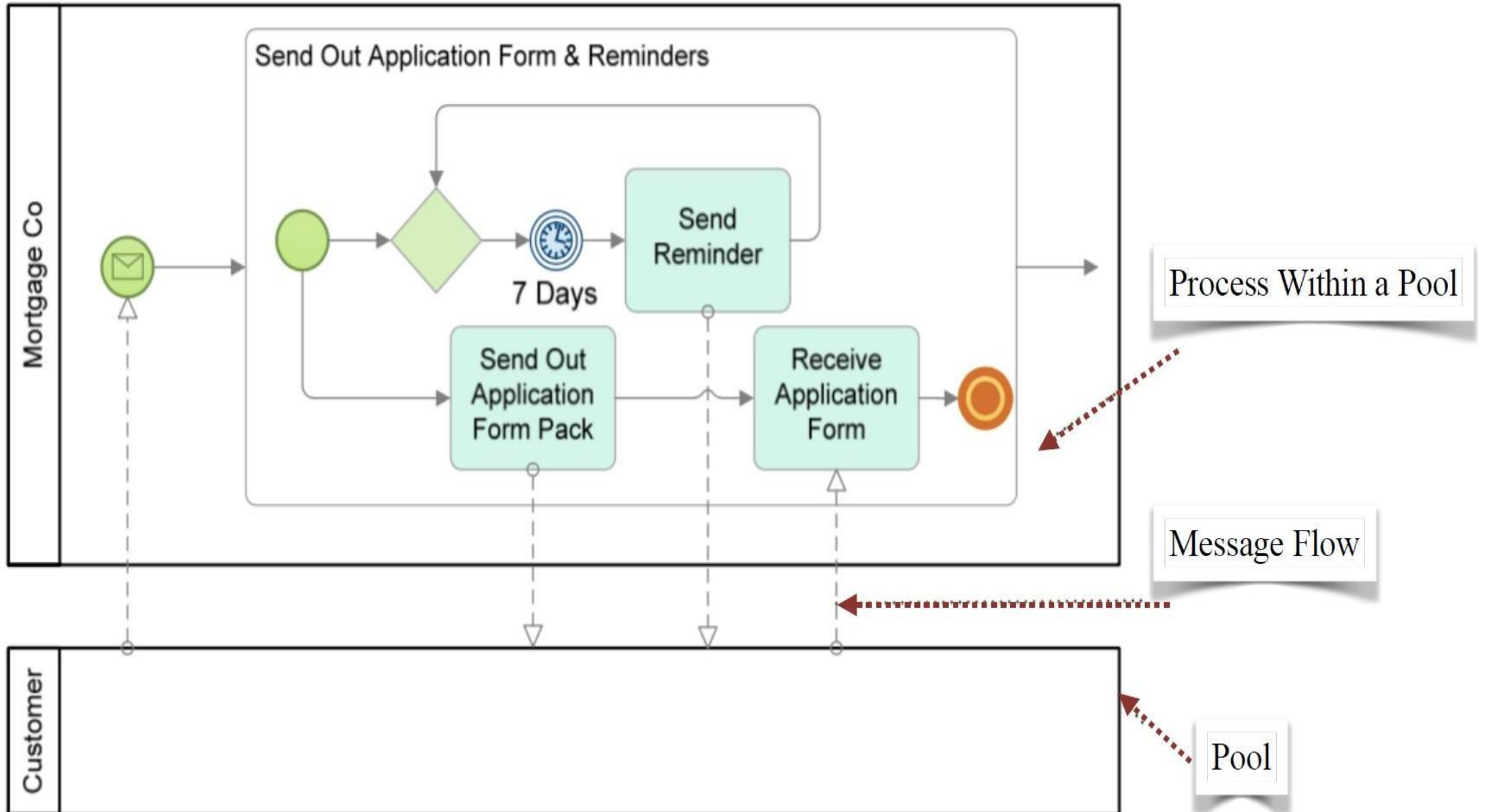
- Swim lanes are just used for visualization purposes.
- With the help of swim lanes, one can easily organize the activities of a business.
- Swim lanes are represented using two objects: the Lane and the Pool.
- Pool: Scope/Boundary and External Actors too
- Lanes: Allow us to associate the tasks of the internal actor



3.3.1.1 Swim Lanes: Pools

- Pools represent a process or a primary participant who carry the process in an interactive (B2B) Business Process Diagram.
- A pool may be a Black Box or a White Box (containing a process).
- Interaction between pools is handled through **Message Flow**.
- Sequence flow cannot cross the boundary of a pool.

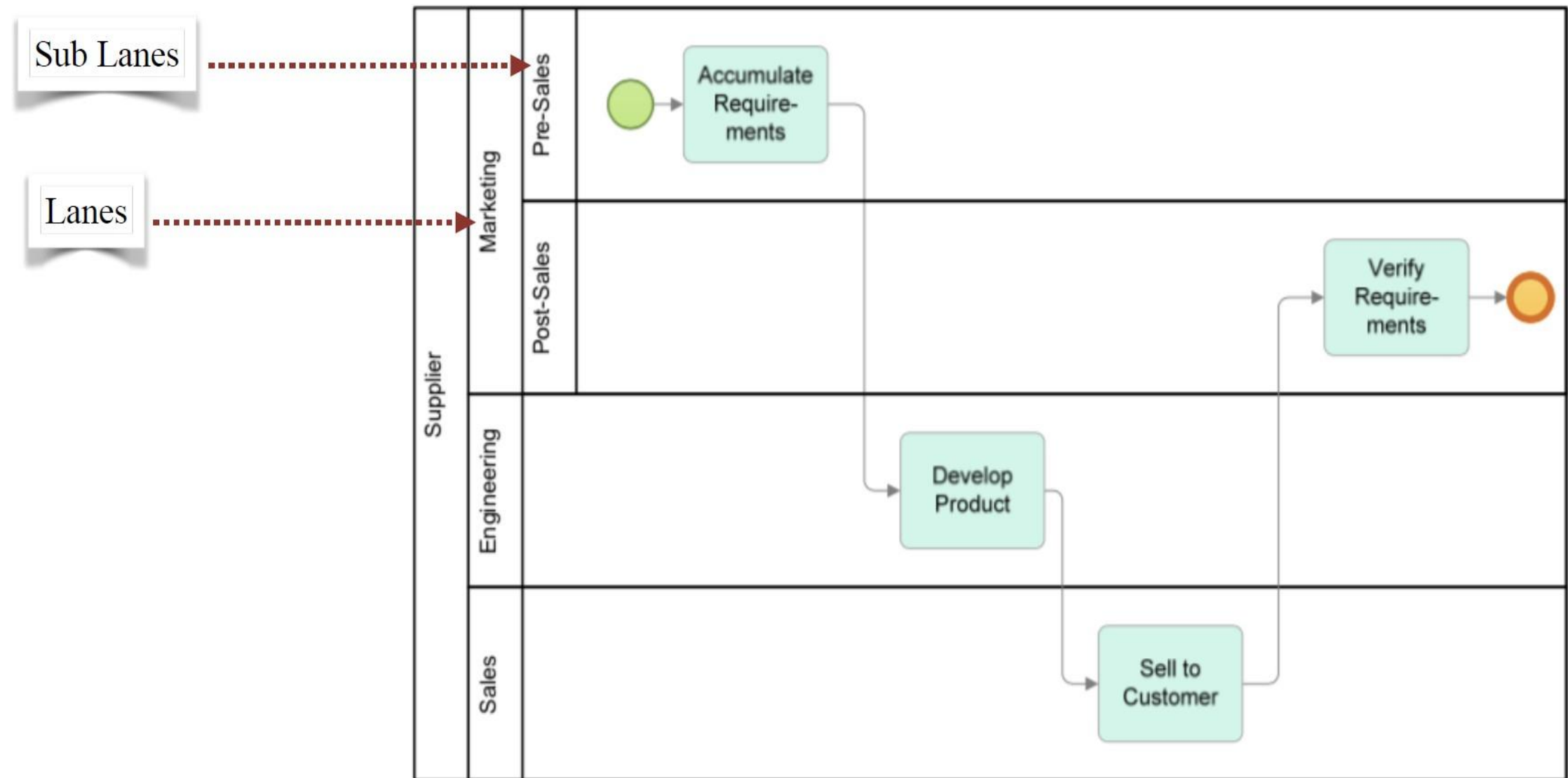
Swim Lanes: Pools - Example



3.3.1.2 Swim Lanes: Lanes

- Lanes represent participants/sub participants for the objects within a Pool.
- They often represent organization roles (e.g. manager) but can represent any desired process characteristic.
- Sequence flow can cross lane boundaries.

Swim Lanes: Lanes - Example

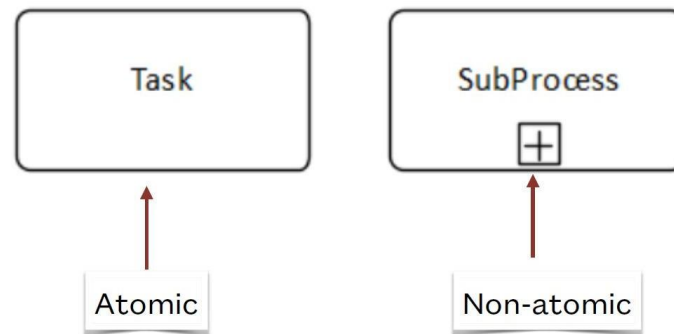


3.3.2 Flow Objects

- Flow objects consist of various shapes that represent the **core elements** of business process representation, which characterize the behavior of a business process. To represent the flow of the object, three elements are used: **Activities, Events and Gateways**.

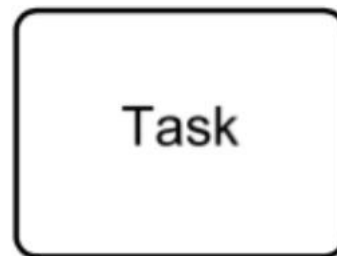
3.3.2.1 Flow Objects: Activity

- Activities represent the tasks or work that is being performed within a business process. The shape that represents the activities in a BPMN is a **round-cornered rectangle**.
- Can have multiple incoming sequence flows.
- If the activity does have an incoming sequence flow, then it must be instantiated when the Process is instantiated.
- An Activity is a work that is performed within a Business Process. An Activity can be atomic or non-atomic (compound).
- There are various types of activities. Such as Task, Sub-Process.



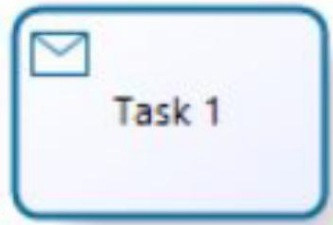
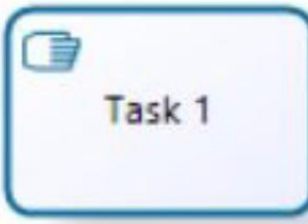
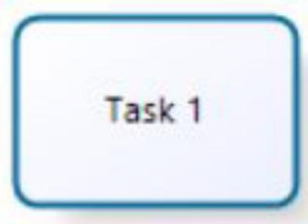
Activities: Tasks

- A Task is used when the work in the Process cannot be broken down to a finer level of detail. Atomic in nature.
- Generally, an end-user and/or applications are used to perform the Task when it is executed.
- Various types of tasks are provided by BPMN, such as the None, Manual **Task**, **Receive Task**, **Script Task**, **Send Task**, **Service Task**, **User Task**, **Business Rule Task**.



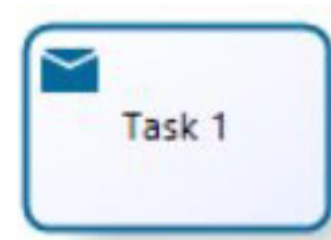
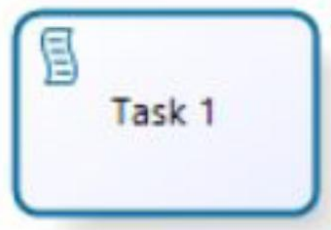
Activities: Tasks (Continued)

- **None** - Generic Task. Single Unit of work that is not or cannot be broken down to a further level of business process detail without diagramming the steps in a procedure.
- **Manual** - A Manual Task is a task that is expected to be performed without the aid of any business process execution engine or any application. An example of this could be a telephone technician installing a telephone at a customer location.
- **Receive** - A Receive Task is a simple task that is designed to wait for a Message to arrive from an external Participant (relative to the Process). Once the message has been received, the Task is completed.



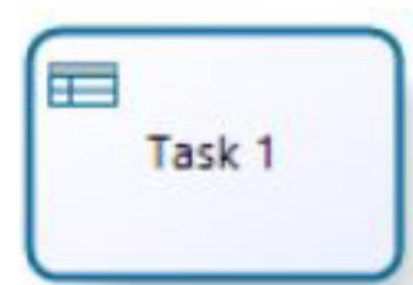
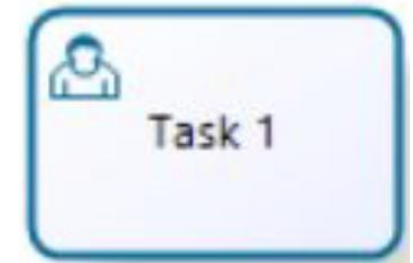
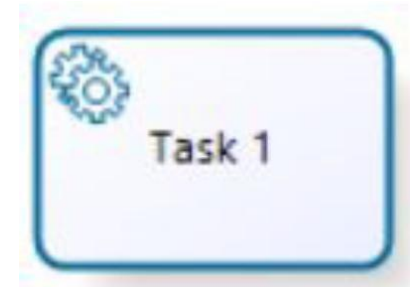
Activities: Tasks (Continued)

- **Script** - A Script Task is executed by a business process engine. The modeler or implementer defines as the script in a language that the engine can interpret. When the task is ready to start, the engine will execute the script. When the script is completed, the Task will also be completed.
- **Send** - A send task is a simple one that is designed to send a message to external participants (relative to the process). Once the message has been sent, the task is completed.



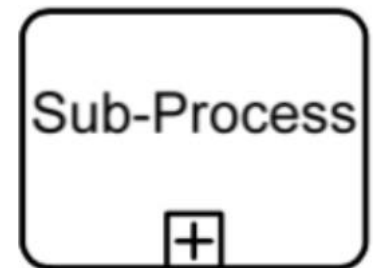
Activities: Tasks (Continued)

- **Service** - A Service Task is a task that uses some sort of service, which could be a Web service or an automated application.
- **User** - A User Task is a typical “workflow” task where a human performer reforms the task with the assistance of a software application and is scheduled through a task list manager of some sort.
- **Business Rule** - A Business Rule Task provides a mechanism for the Process to provide input to a Business Rules Engine and to get the output of calculations that the Business Rules Engine might provide. The input output specification of the task will allow the Process to send data to and receive data from the Business Rules Engine.



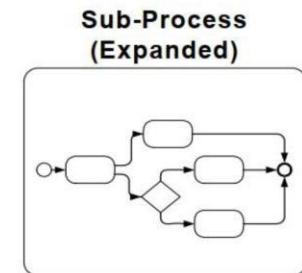
Activities: Sub Process

- When capturing complex business processes, the resulting process model may be too large to be understood at once.
- To improve its readability, we can simplify the process by hiding certain parts within a sub-process.
- In order to use a sub-process, we first need to identify groups of related activities .
- Sub-process is Activity with an internal structure containing activities, gateways, events and sequence flows.



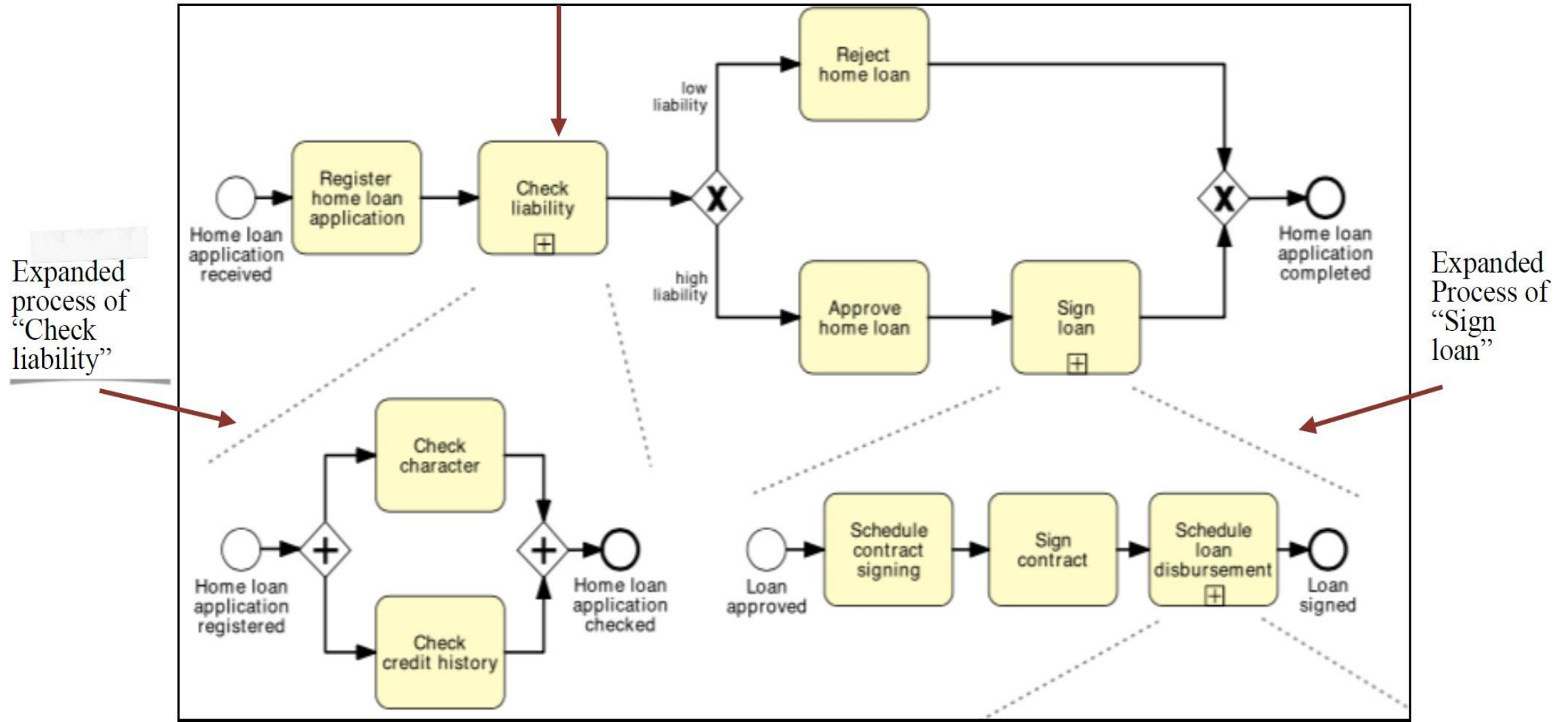
Activities: Sub Process (Continued)

- Once we have identified the boundaries of the sub-process , we can simplify the model by hiding the content of it.
- We indicate that this activity hides a sub-process by marking it with a small square with a plus sign (+) inside.
- This operation is called collapsing a sub-process.
- A sub-process which hides its internal steps is called collapsed sub-process, as opposed to an expanded sub-process which shows its internal steps.



Activities: Sub Process (Continued)

Collapsed Sub-process



3.3.2.2 Flow Objects: Events

- An Event is something that “**happens**” during the course of a Process or a Choreography.
- These events affect the flow of the model and usually have a cause (Trigger) or an impact (Result).
- Events are circles with open centers to allow internal markers to differentiate different Triggers or Results.
- There are three types of Events, based on when they affect the flow: **Start, Intermediate, End**



Start



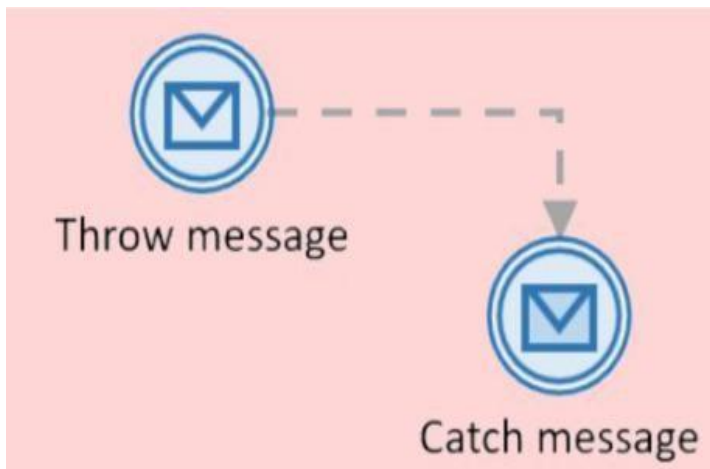
Intermediate



End

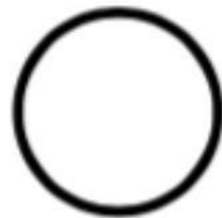
3.3.2.2 Flow Objects: Events (Continued)

- Within those three types, events come in two categories;
 - Events that **catch** a trigger. All the start events and some intermediate events are catching events.
 - Events that **throw** a result. All end events and some intermediate events are throwing events that may eventually be caught by another event.



Events: Start Event

- Start Event indicates where a particular Process or Choreography will start.
- A Start Event is a small, open circle with a single, thin lined boundary.

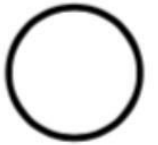


Start Event

Events: Start Event (Continued)

Core set of Start Events:

- **None** - No trigger is defined.
- **Timer** - The trigger is specific date and time, or a regular date-time cycle (eg: the first Friday of the month at 8 am).
- **Message** - The trigger is a message arrives from another business entity or role (participant). Eg: a client requests a review of their account.
- **Signal** - The trigger is a signal broadcast from another Process. Eg: a Process broadcasts a change in the prevailing Interest Rate, triggering any number of other processes to start as a result.



Events: Start Event (Continued)

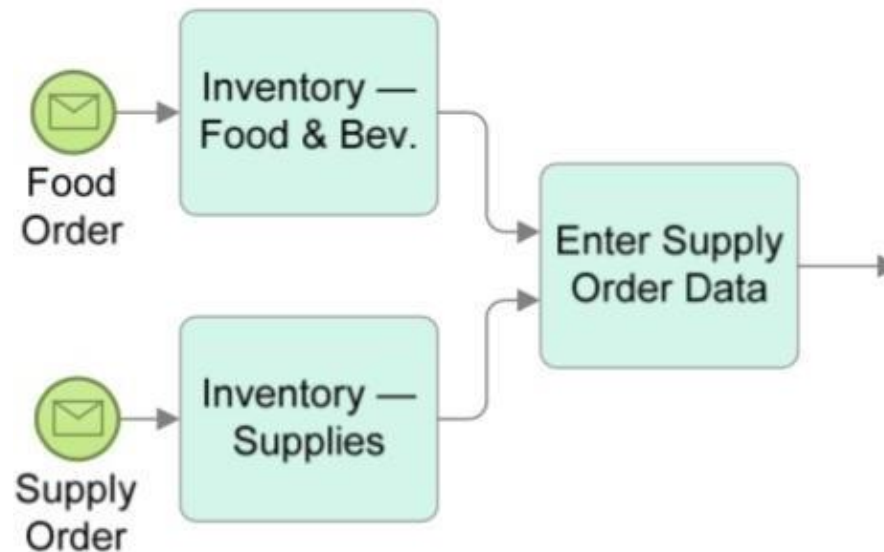
- **Conditional** – This type of event is triggered when a condition such as “stock level < 500”, or “Temperature above 300C” become true.
- **Multiple** – This means that there are multiple ways of triggering the process. Only one of them is required.
- **Parallel Multiple** – Multiple triggers required before the process can be instantiated. All of the types of triggers that are listen in the start event must be triggered before the process is instantiated.



Events: Start Event (Continued)

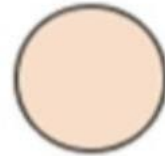
Modeling with More than One Start Event

Most Processes have a single Start Event. However, it is possible to include more than one Start Event within a single Process .

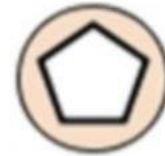


Events: Start Event (Continued)

All Types of Start Events



None



Multiple



Conditional



Message



Timer



Signal



Error



Escalation



Copression

Events: End Event

- End Event indicates where a Process or Choreography will end.
- In terms of Sequence Flows, the End Event ends the flow of the Process, and thus, will not have any outgoing Sequence Flows.
- No Sequence Flow can connect from an End Event.
- An End Event is a small, open circle with a single, thick lined boundary.



End Event

Events: End Event (Continued)

- Unlike Start events its best practice to have more than one
- Each end event documents the state of the process at the end
- In a process (or sub-process) every parallel path must reach an end event for the process to complete
- A sub process completes when all its parallel paths have reached an end event
- It is not necessary, and in fact bad practice to join the parallel paths immediately before an end event

Events: End Event (Continued)

Core set of End Events:

- **None** - No result is defined. Name after the condition describing the final process state. Eg-Leave approved, Application Rejected
- **Message** - Process ends by sending a message. Eg: Goods Dispatched, Notice Sent.
- **Signal** - Defines a “broadcast” Event that any other Process can see and react to.



Events: End Event (Continued)

- **Error** – A named error should be generated. All currently active threads in the particular sub-process are terminated as a result. Immediately terminates that process or sub-process, even if there are parallel paths still active.
- **Cancel** – Cancel the transaction and triggers a named intermediate event
- **Terminate** - Stops all Process Activities, even if there are ongoing in other Threads (Parallel Paths). Kills all incomplete tasks in process or sub process.



Events: End Event (Continued)

All Types of End Events



None
(not defined)



Escalation



Signal



Message



Cancel



Terminate



Error



Compensation



Multiple

Events: Intermediate Events

- Intermediate Events occur between a Start Event and an End Event.
- They will affect the flow of the Process or Choreography, but will not start or (directly) terminate the Process.
- An Intermediate Event is a small, open circle with a double, thin lined boundary.



Intermediate Event

Events: Intermediate Event (Continued)

Core set of Intermediate Events:

- **None** - An Intermediate Event without a trigger is known as a None Intermediate Event.
- **Timer** - Timer Intermediate Events can only catch. They add time-based dependencies within a Process and are either inserted into the Sequence Flow to create a delay or attached to the boundary of an Activity to create a deadline or “time-out” condition.
- **Message** - The trigger is a message. One type of Message Intermediate Event is a throwing Event that sends a message. The other type of Message Intermediate Event catches.
- **Signal** - The trigger is a signal that is broadcast or received.



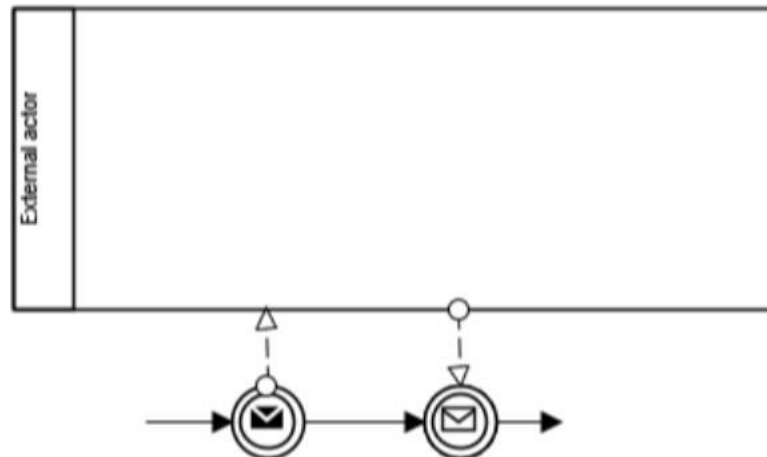
Events: Intermediate Event (Continued)

All Types of Intermediate Events

	Catching	Throwing
None		
Timer		
Message		
Signal		
	Catching	Throwing
Error		
Cancel		
Compensation		
Conditional		
Link		
Multiple		

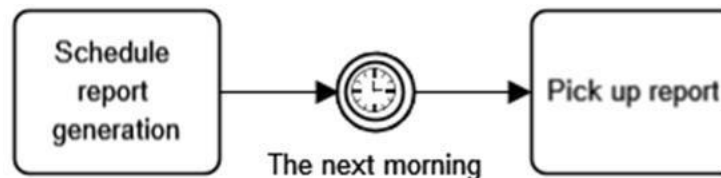
Events: Intermediate Events (Continued)

- When modeling a business process that contains a pause related to waiting for a response from an external actor, it is recommended to present the exchange with the external actor with an throwing message intermediate event.
- This intermediate event initiates the exchange, makes a request to the external actor and is followed by an catching message intermediate event that holds the token until the external actor's response is received.



Events: Intermediate Events (Continued)

- For the timer intermediate event, it is important to specify the length of the pause.
- The figure below contains a timer intermediate event that illustrates a pause between the activities immediately before and after the event.
- In this example, the report is retrieved the morning after report generation is scheduled. After the first activity, the timer event pauses the token. It will be released when the time condition is reached, which is the next morning:



Events: Intermediate Event (Continued)

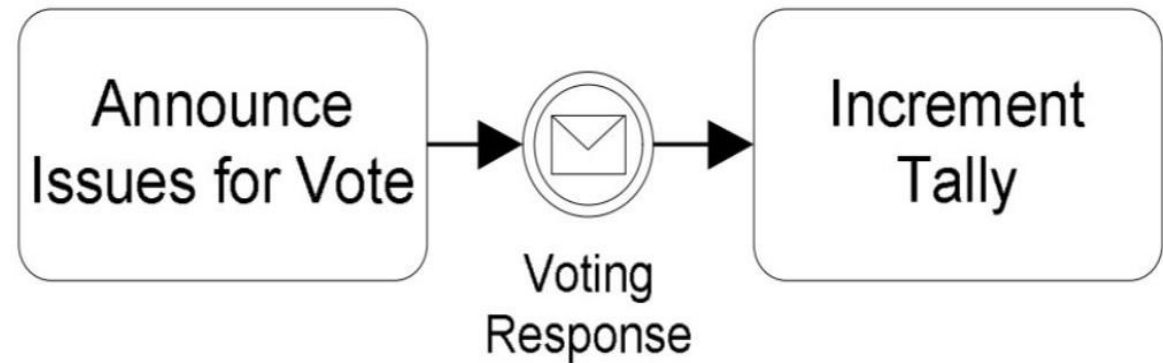
Boundary Vs In-flow:

- Can be attached to the boundary of a sub-process, or embedded in the sequence flow.
- All boundary events must be of the catch variety.
- Error events can be caught only on the boundary.
- Link events can only be caught in the flow.
- The effect of throwing a boundary event in a subprocess is to interrupt the sub-process and immediately transfer control to the boundary (catch) event.

Events: Intermediate Event (Continued)

In-flow:

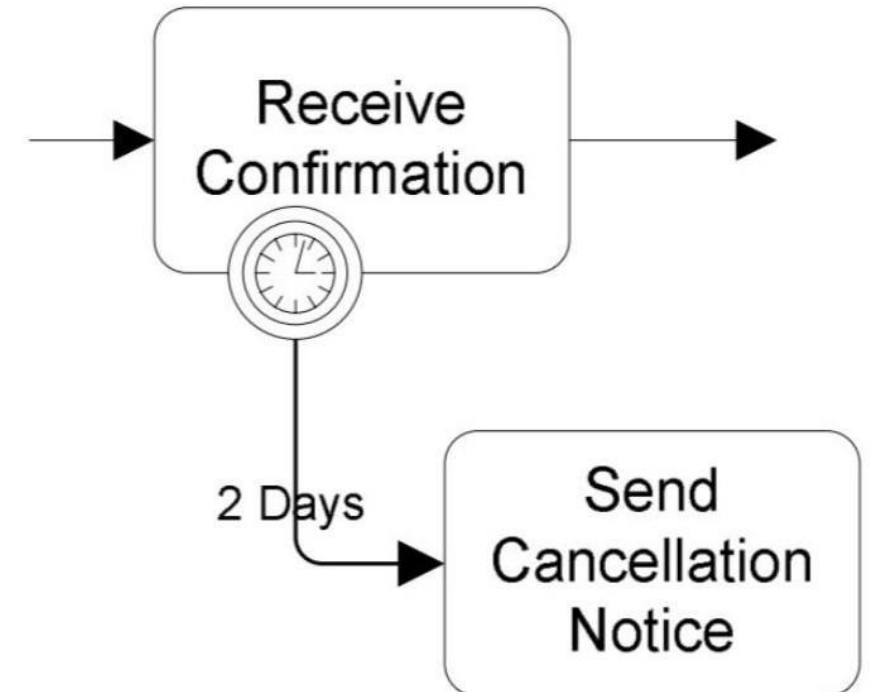
- Events that are placed within the process flow represent things that happen during the normal operations of the process
- They can represent the response to the Event (i.e., the receipt of a message)
- They can represent the creation of the Event (i.e., the sending of a message)



Events: Intermediate Event (Continued)

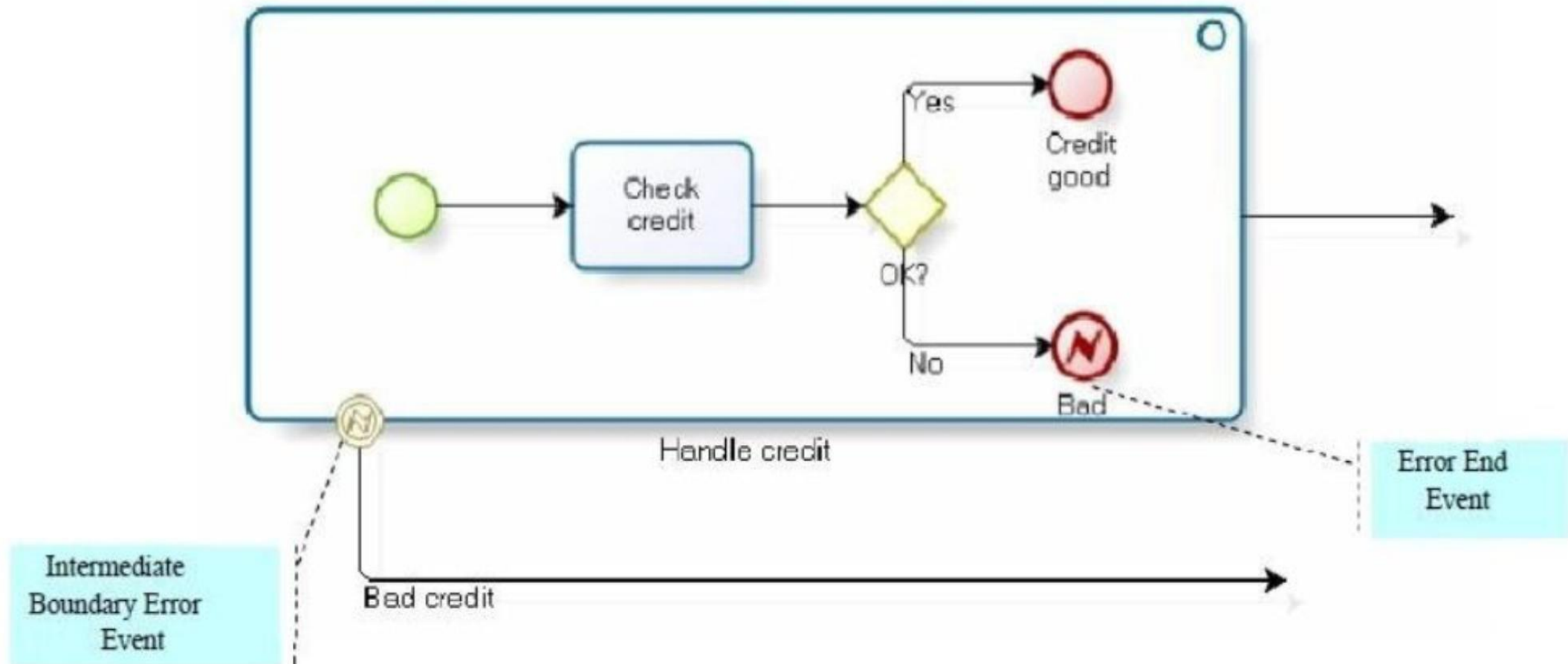
Boundary:

- Events that are attached to the boundary of an activity indicate that the activity should be interrupted when the Event is triggered.
- They can be attached to either Tasks or Sub-Processes.
- They are used for error handling, exception handling, and compensation.



Events: Intermediate Event (Continued)

Boundary Vs In-flow:



Events: Starting an Event

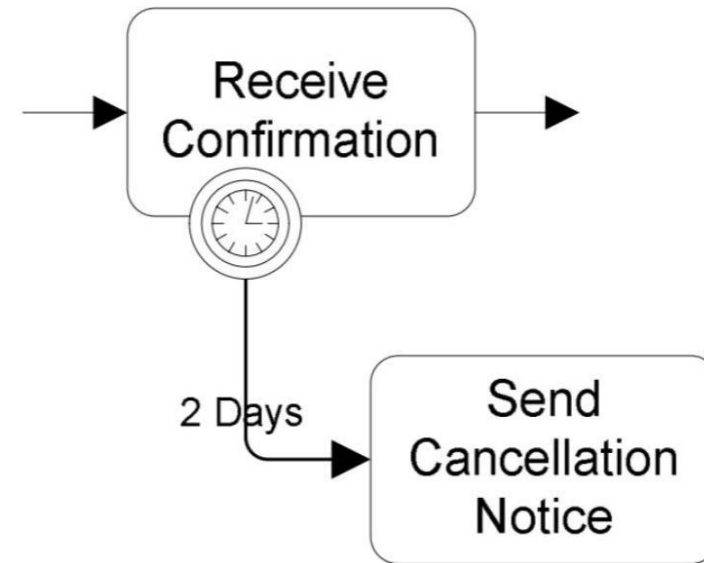
Boundary Vs In-flow:

- Previous activity must complete before the activity following can begin.
- Delays
 - ◆ Event Delays
 - Modeled as an Intermediate Catch Event.
 - Activity may require a message from an external participant. Modelled as an Intermediate **Catch Message Event**.
 - An Activity may not begin until a certain time has elapsed. Modeled by an Intermediate **Timer Event**.
 - ◆ Resource Delays
 - An activity might be delayed until a resource becomes available. NOT modeled in BPMN
 - As an example a user activity will require a human actor to perform it.

Timers



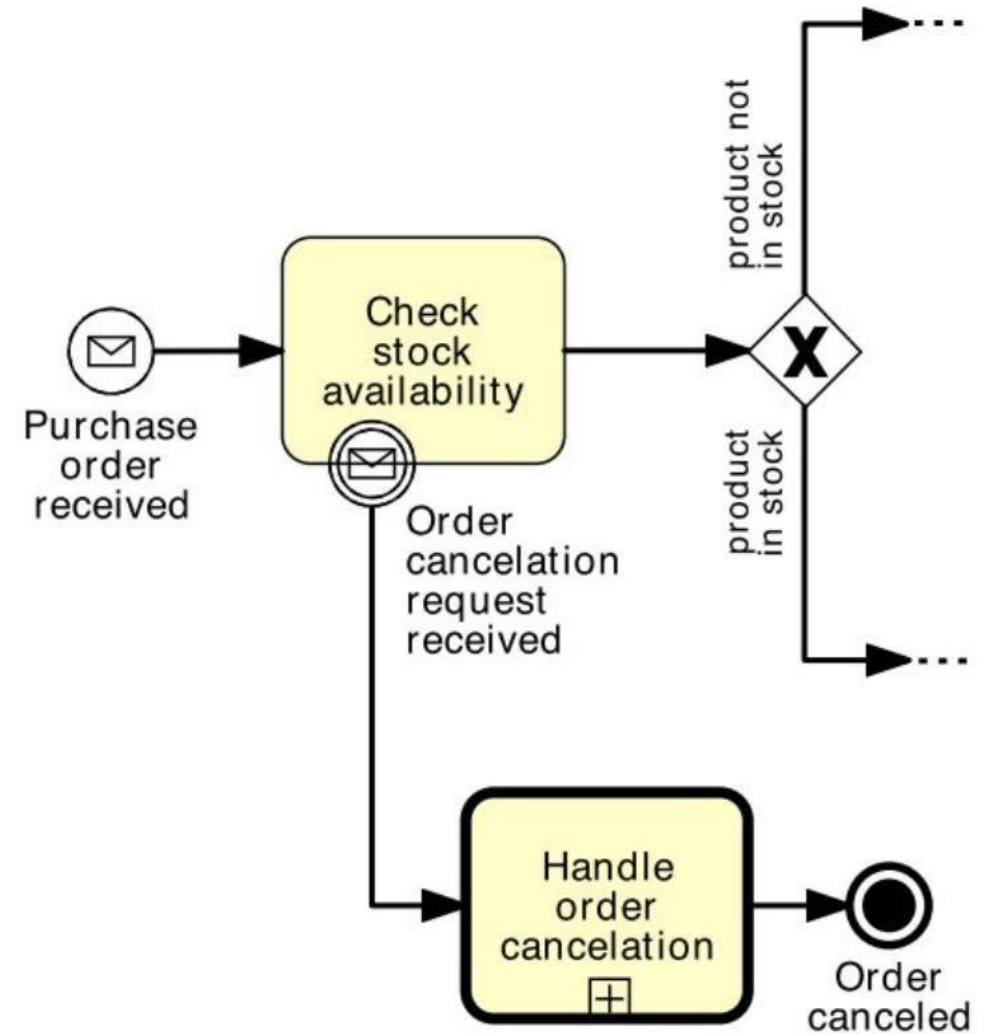
As a Delay



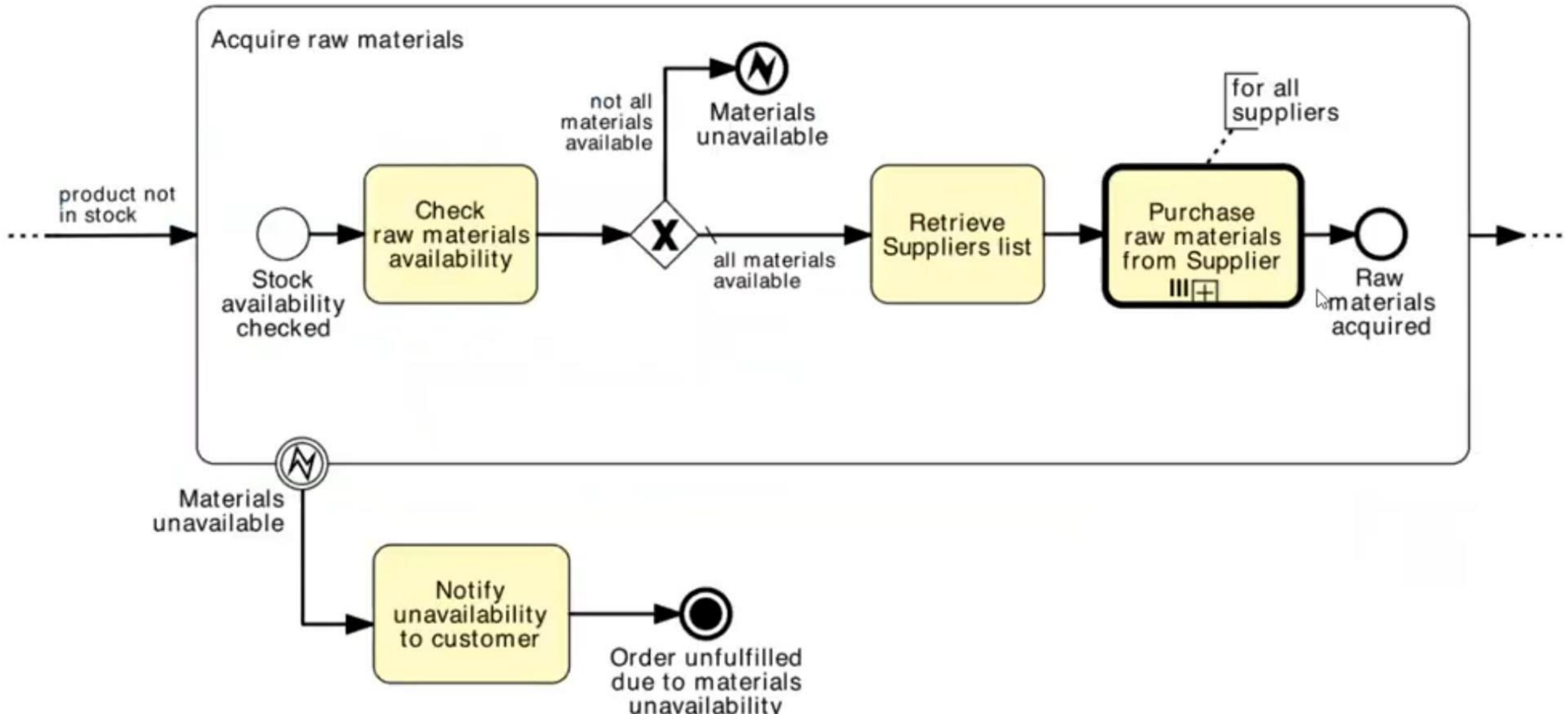
As a Exception

Interrupting Events

- Intermediate Events attached to the boundary of an activity represent triggers that can interrupt the activity.
- All work within the activity will be stopped and flow will proceed from the Event.
- Timer, Errors, Messages, etc. can be Triggers.

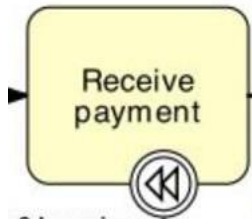


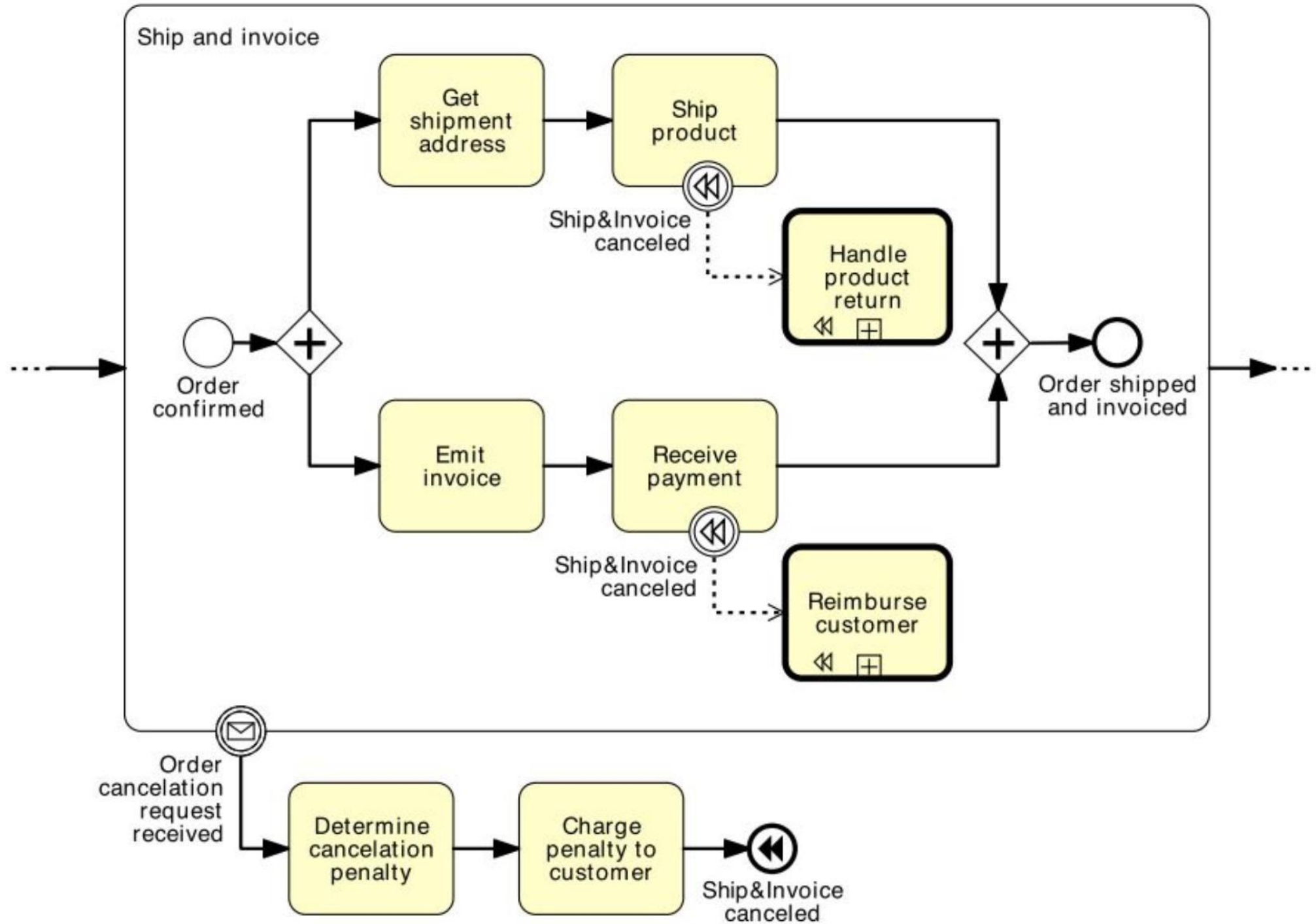
Exception Handling



Compensation

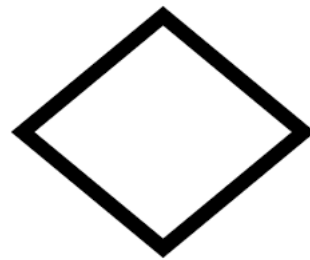
- Restore the process to a business state close to the one before starting the sub-process that was interrupted.
- A compensation handler is made up of,
 - ◆ throwing compensate event (an event marked with a rewind symbol),
 - ◆ a catching intermediate compensate event and
 - ◆ a compensation activity.
- Activities used for compensate (with marker) are outside normal flow and are Associated normal activities.





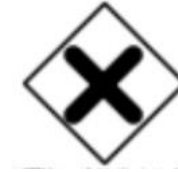
3.3.2.3 Flow Objects: Gateways

- Gateways are used to handle the forking and joining of paths within the business process.
- The main aim of gateways is to control the flow of a business process.
- The diamond shape represents a gateway.
- The types of gateways available are **Exclusive Gateway, Inclusive Gateway, and Parallel Gateway.**

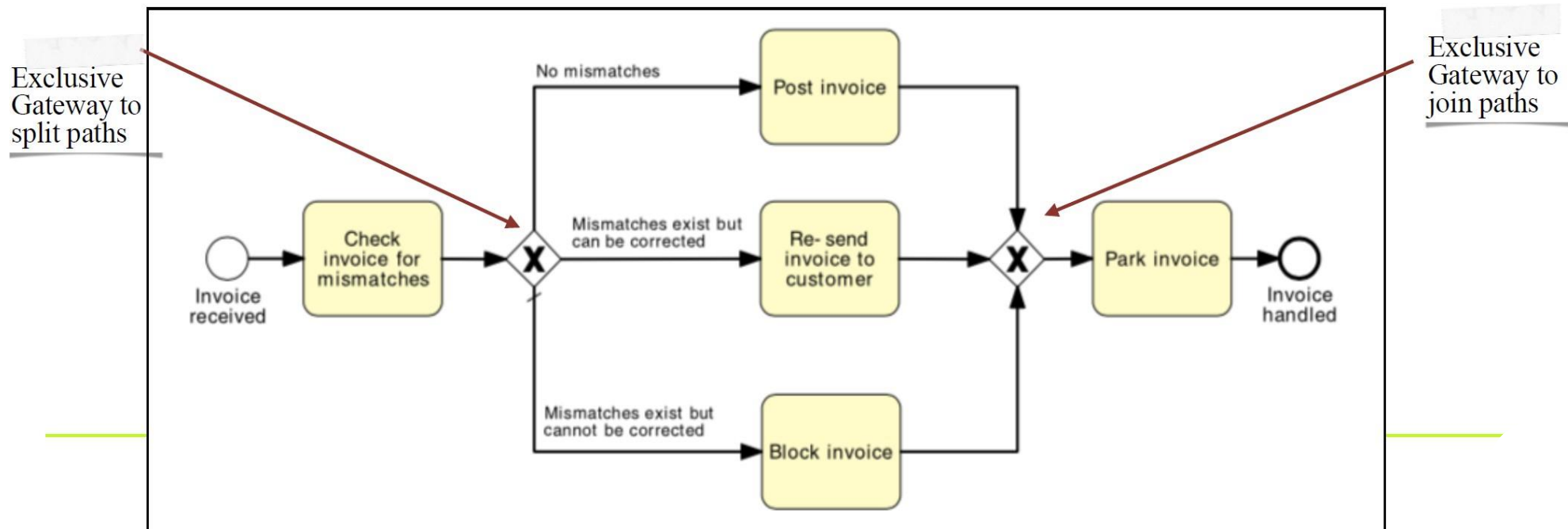


Gateway

Gateways: Exclusive Gateway



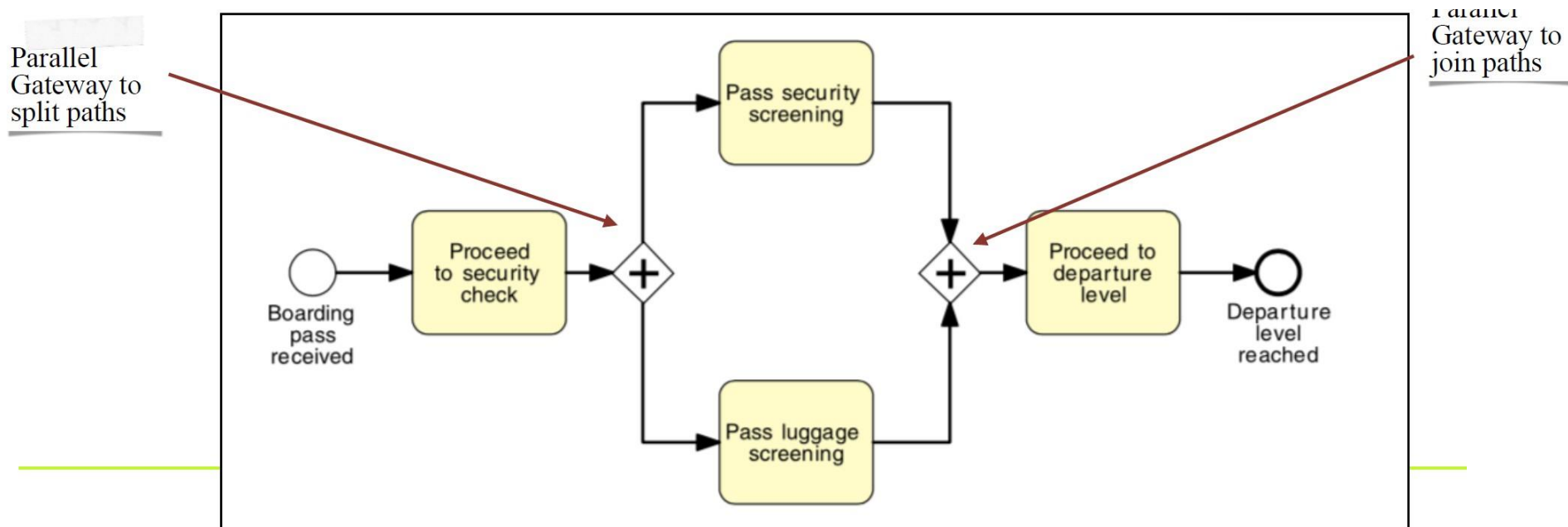
- We can use an exclusive gateway when we want to proceed with one path from the multiple paths defined.
- So, we can compare the exclusive gateway to an if-else statement of the programming concept.



Gateways: Parallel Gateways



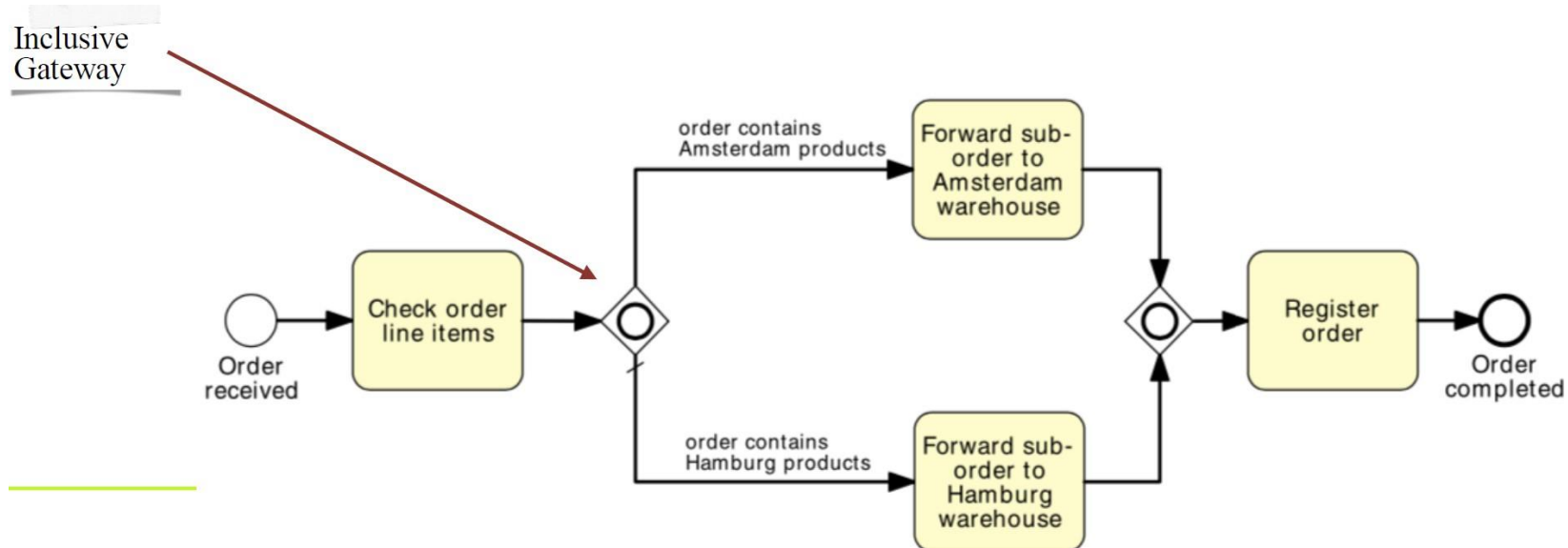
- A Parallel Gateway is used to synchronize (combine) parallel flows and to create parallel flows.
- When two or more activities **do not have any order dependencies** on each other, they can be executed concurrently or in parallel.



Gateways: Inclusive Gateways

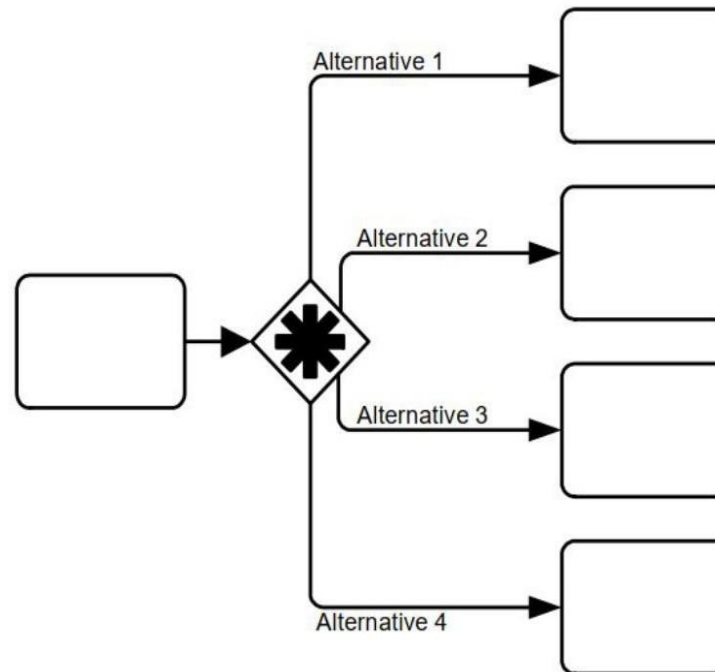


- Inclusive Gateway (Inclusive Decision) can be used to create an alternative but also parallel paths within a Process flow.
- An inclusive gateway can be used when we want to execute the paths for which the conditions specified for them are satisfied. Inclusive gateway is the combination of AND and XOR gateways.



Gateways: Complex Gateways

- Complex Gateways are Decisions where there is more advanced definitions of behaviors can be defined.
- The asterisk marker is used to identify this gateway
- Complex behavior can be defined for both the merging and splitting behavior



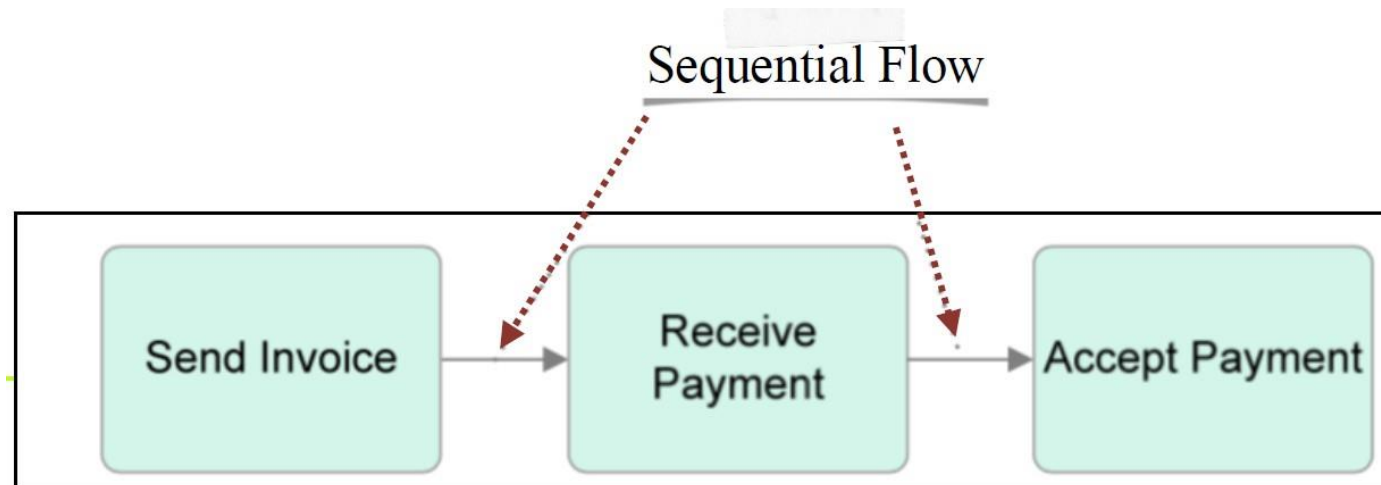
3.3.3 Connecting Objects

- Connecting objects are used to represent a basic framework of a business process structure.
- Each flow object can be connected using the connecting objects.
- These connecting objects can be represented using the following types: **Sequence Flow, Message Flow, and Association.**

Connecting Objects: Sequential Flow



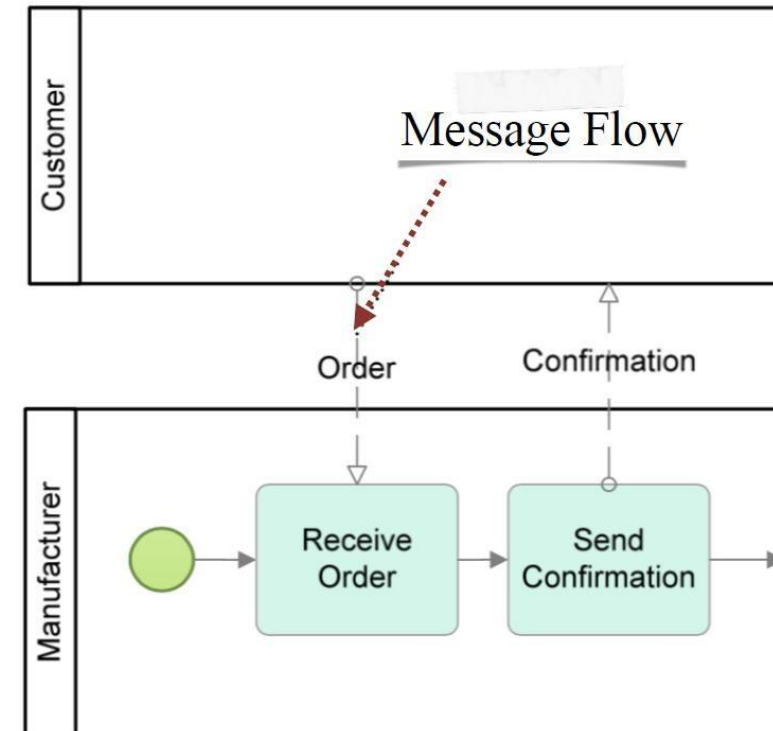
- A Sequence Flow is used to show the order that activities will be performed in a process.
- Defines the order of flow objects in a Process (Activities, Events, and Gateways).
- A Sequence Flow CANNOT cross a Sub Process Boundary or a Pool Boundary.



Connecting Objects: Message Flow



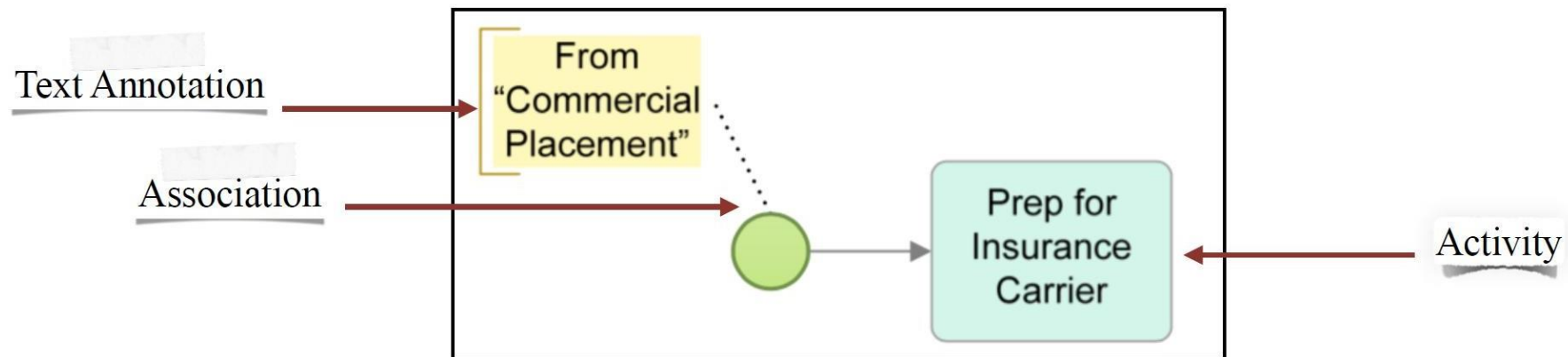
- The Flow of Messages between two Participants that are prepared to send and receive them.
- A Message Flow MUST connect two separate Pools.
- They connect either to the Pool boundary or to Flow Objects within the Pool boundary.
- They MUST NOT connect two objects within the same Pool.



Connecting Objects: Association



- An Association is used to associate artifacts with flow objects.
- An Association will link one diagram object with another diagram object.



Default Flow

- For Data-Based Exclusive Gateways or Inclusive Gateways, one type of flow is the Default condition flow.
- This flow will be used only if all the other outgoing conditional flow is not true at runtime.
- Normal Flow does not include exception flow or compensation flow.



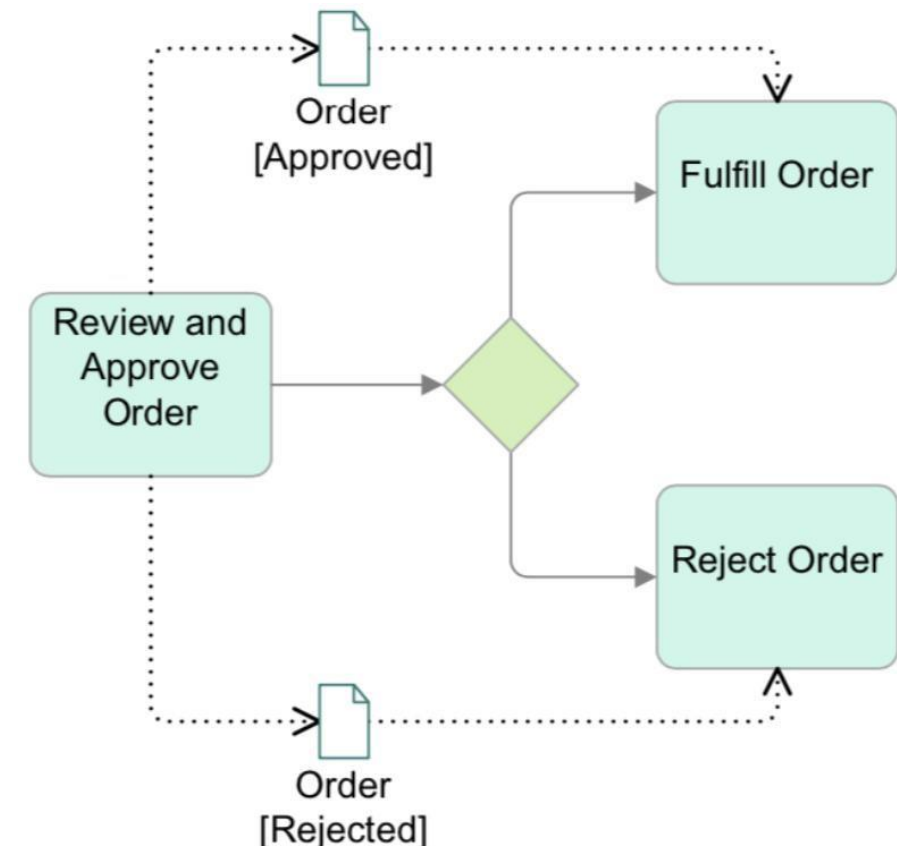
3.3.4 Artifacts

- Artifacts are used to provide additional information in BPMN diagrams .
- There are two types of artifacts: **Data Object , Group and Annotation.**

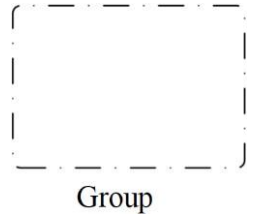
Artifacts: Data Object



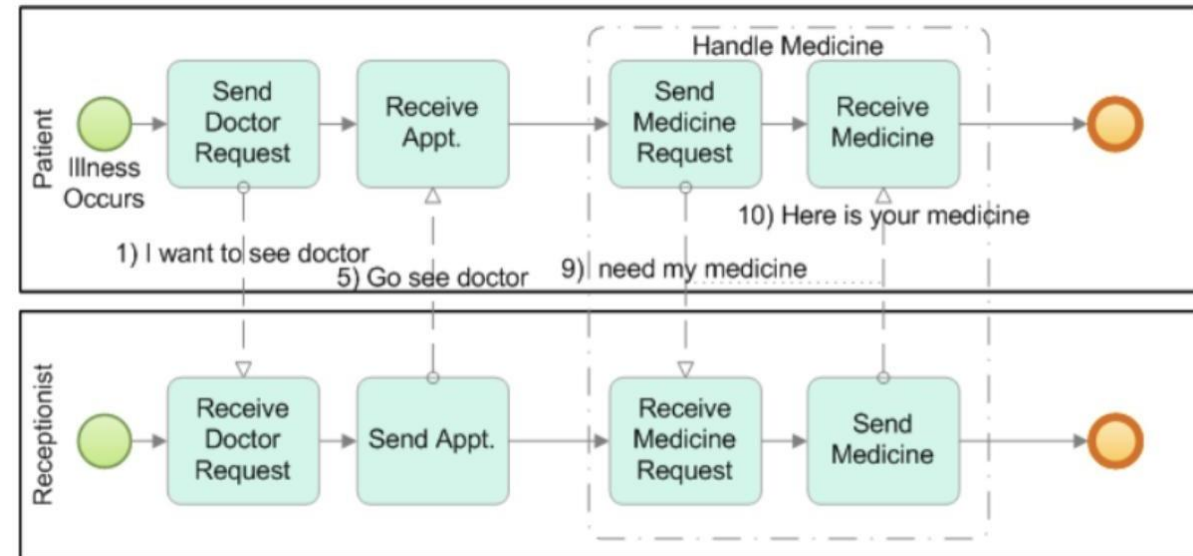
- Data Objects usually define the inputs and outputs of activities.
- With the help of this artifact, one can get an idea regarding the data required or produced within an activity.
- Data Objects also have “states” that depict how the object (document) is updated within the Process. Eg: Order [Approved], Order [Rejected].
- It is represented by an image of a piece of paper folded at the top-right corner.



Artifacts: Group



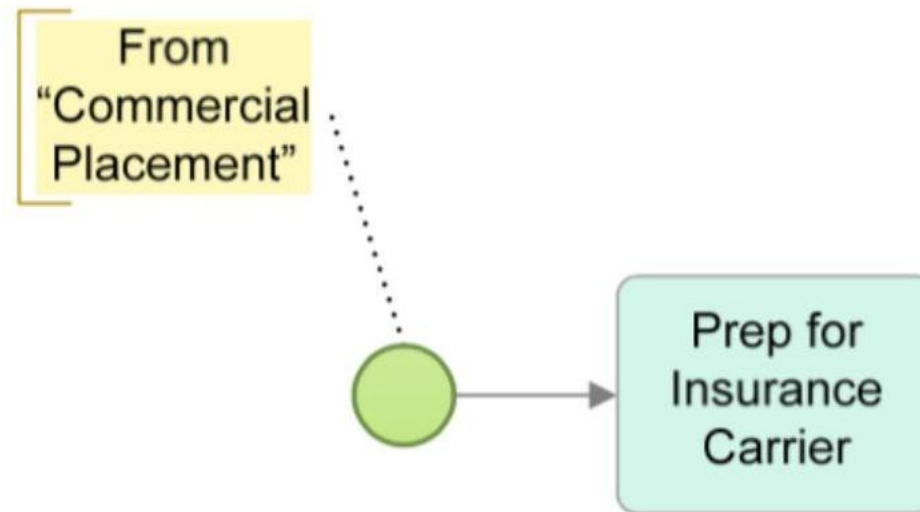
- A Group is a grouping of graphical elements that are within the same Category.
- This type of grouping does not affect the Sequence Flows within the Group.
- The Category name appears on the diagram as the group label. Categories can be used for documentation or analysis purposes.
- A group is denoted by a long dash and a dot make up the line.



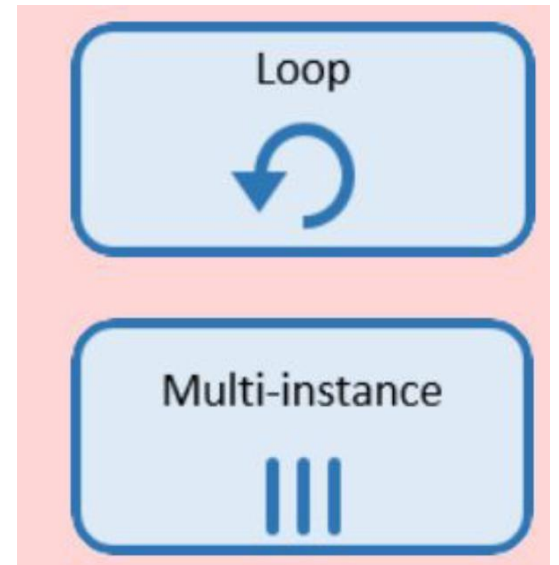
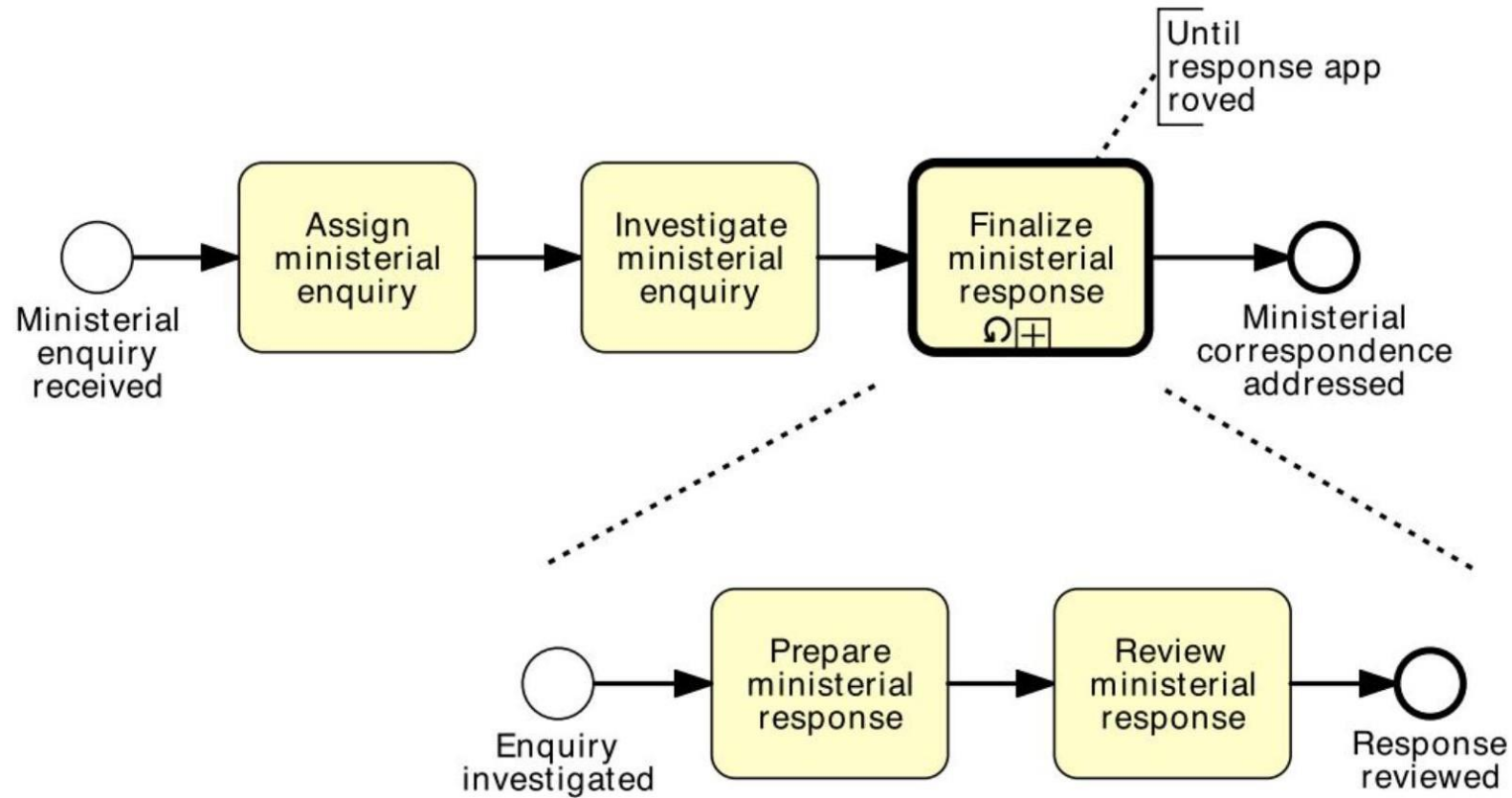
Artifacts: Annotation



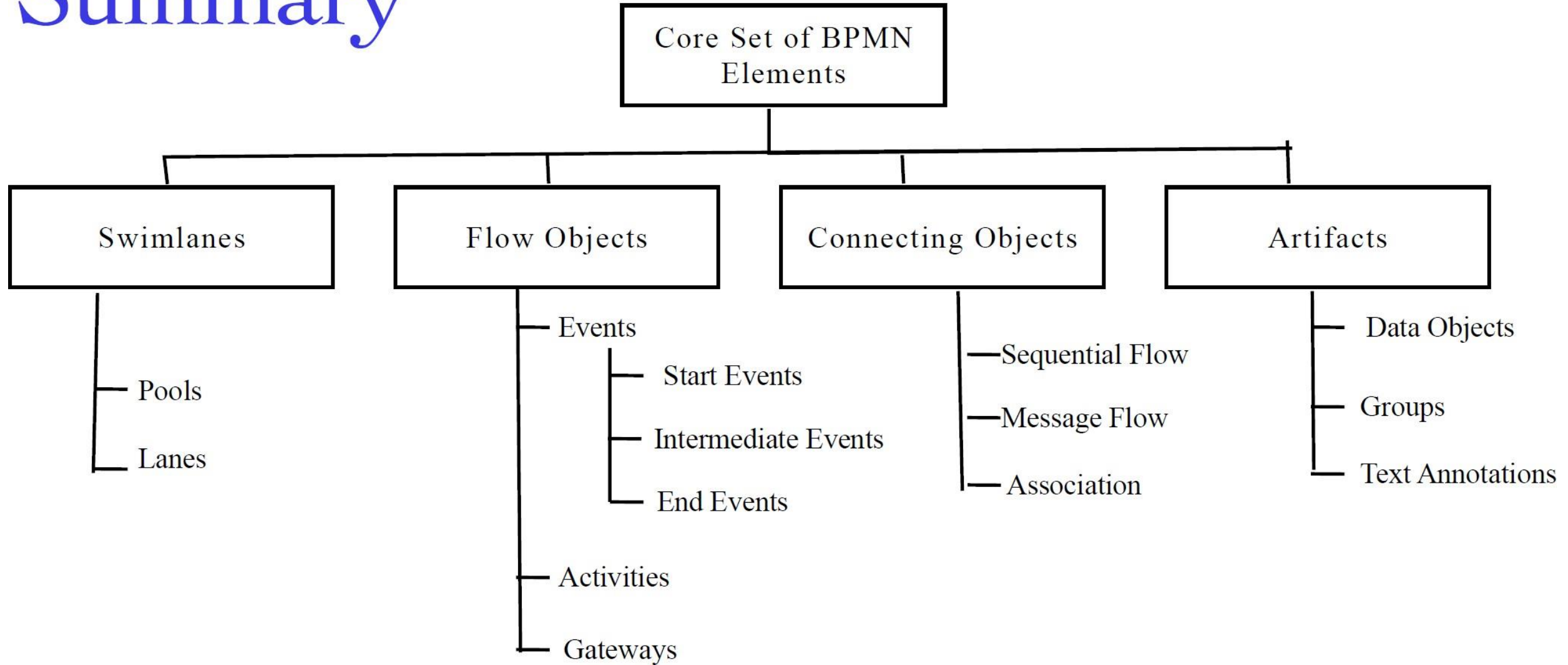
- The annotation artifact is used to add some additional comments to the business process.
- The text for the Text Annotation is accompanied by an open box that can appear on either side of the text



Looping



Summary

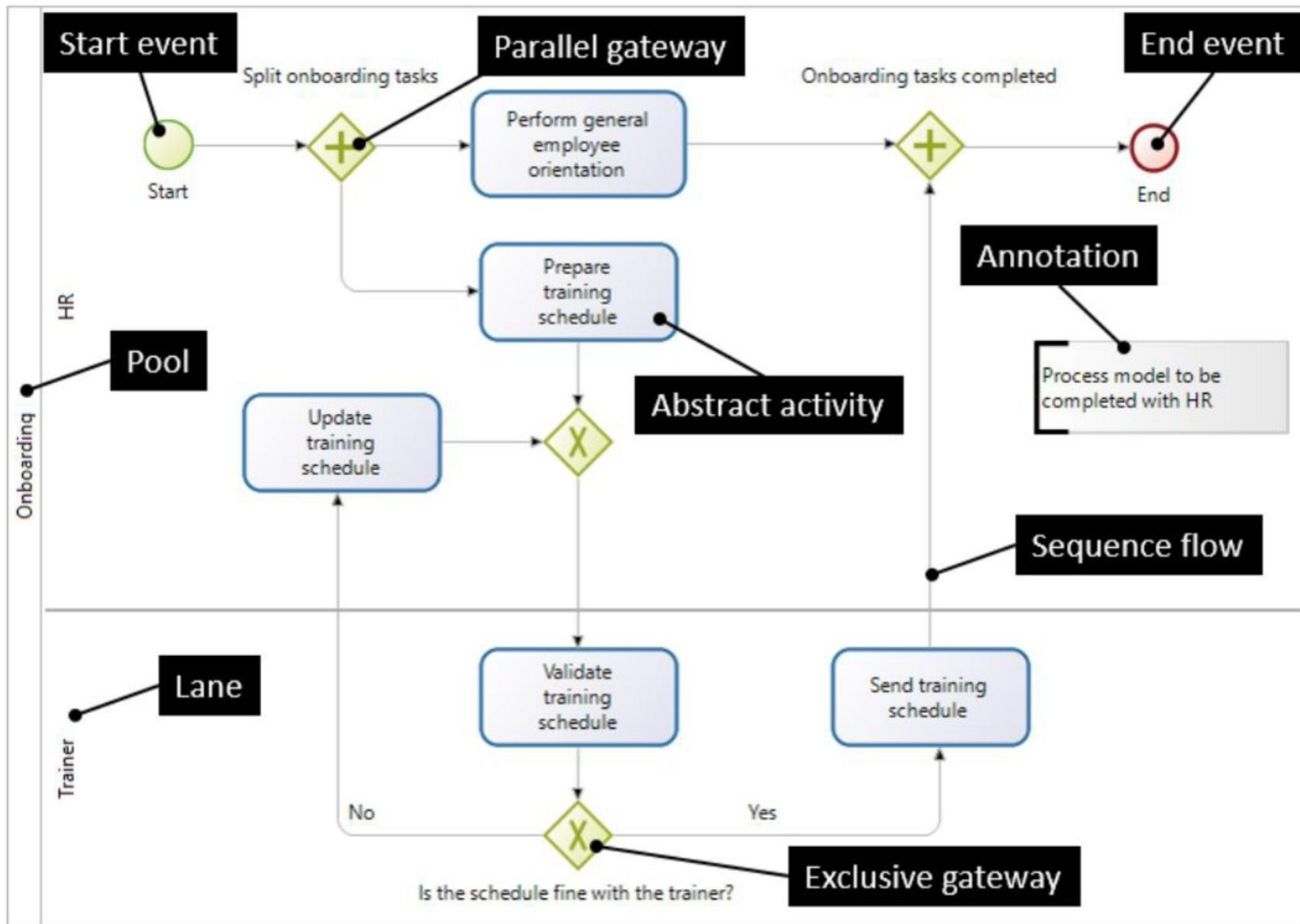


General Guidelines

- A process is chronological. Accurate models should be oriented on a timeline.
- Processes generally begin with triggering events, and work their way through to significant business results.
- All tasks or activities are assigned to roles that are meaningful to people in the business. Be sure you have captured all relevant roles, which may sometimes be outside of the client's company
- A complete model should display how objects or data (or both) are transferred and where they are going
- A process can be modeled in a hierarchical fashion (e.g., with Sub-Processes)
- The choices made for decisions, which occur within a process, determine which of all potential paths will be taken.

General Guidelines (Continued)

- Follow naming conventions.
 - ◆ For example, all activity names could have the following format
 - verb + (adjective/descriptor) + noun
 - example: “Verify Account
 - ◆ Avoid redundancy in naming, e.g., do not include the word Process in the Process names or the words Task or Activity in Task names
 - ◆ To help with report outputs, names should be 32 characters or less
- Establish a set of standard nouns, verbs, and acronyms that are used for naming objects.
- Establish standards for versioning methods associated at the process model and artifact level to provide requirement traceability.



Thank You!